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RESEARCH-ARTICLE

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Breathing exercises reduce stress and improve overall mental well-being. There are various types of breathing exercises. Performing the exercises correctly may give the best outcome and doing it in wrong ways can sometimes have adverse effects. Providing real-time biofeedback can greatly improve the user experience in doing the right exercises in the right ways. In this paper, we present methods to passively track breathing biomarkers in real-time using wireless commodity earbuds and generate feedback on users' breathing performance. We use the earbud's low-power accelerometer to generate a comprehensive set of breathing biomarkers including breathing phase, breathing rate, depth of breathing, and breathing symmetry. We have conducted studies where the subjects performed different types of guided breathing exercises while wearing the earbuds. Our algorithms detect breathing phases with 90.91% F1-score and estimate breathing rate with 95.05% accuracy. We further show that our algorithms can be used to generate biofeedback towards designing engaging smartphone's user interactions that facilitate users to accurately perform various breathing exercises.

CCS Concepts: • **Human-centered computing → Ubiquitous and mobile computing systems and tools**;

Additional Key Words and Phrases: Guided Breathing, Breathing Biofeedback, Mental Well-being, Earbuds

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1 INTRODUCTION

Stress is prevalent in everyday life. Stress has adverse effects on our mental well-being [52], quality of interpersonal relationship [8], and reduces productivity [51]. Repeated, long-term exposure of stress can lead to serious cardio-vascular diseases [49]. Moreover, the ongoing coronavirus (COVID-19) pandemic disrupts our lifestyle and further aggravates stressful situations for people around the world. According to the Stress in America survey conducted by American Psychological Association, nearly half of Americans (49%) report their behavior has been negatively affected due to stress [5]. Most commonly, they report increased tension in their bodies (21%), “snapping” or getting angry very quickly (20%), unexpected mood swings (20%), or screaming or yelling at a loved one (17%) [5].

Breathing exercise is an effective method to reduce stress [24, 41, 44] and improves overall mental well-being [14, 45, 47]. Breathing exercises can be guided by following acoustic [9], haptic [32], or visual instructions [44], or guided by biofeedback where individuals consciously control their own breathing by monitoring the bio-signals [41, 46].

There are different types of breathing techniques including pursed-lip, diaphragmatic, deep breathing with different patterns such as box breathing [2] (4-second inhale, 4-second hold, 4-second exhale, 4-second hold), relaxing breath [54] (4-second inhale, 7-second hold, 8-second exhale), among several others. Despite its direct beneficial impacts, there are some preliminary works on technology-driven interventions that showcase the potential for deep breathing to fail and rather exacerbate stress if not performed properly [3]. These studies show that some people, with no prior exposure to breathing exercises, end up with higher levels of stress. Breathing biomarkers such as respiratory rate [1], breathing phase [46], breathing depth, breathing symmetry (the ratio of inhalation to exhalation) [53] have been evaluated as objective measures of exercise efficacy [16].

However, there is a lack of technologies that can provide those markers as real-time biofeedback during breathing for users to better understand and accurately perform the exercises [3]. In a controlled setting, respiratory inductance plethysmography [15] chest bands have been used as the ‘gold standard’ tool to obtain respiratory rate. These are expensive, uncomfortable, and not suitable for daily use.

To address these limitations, we present BreatheBuddy that uses earbud’s low-power accelerometer sensor to offer fine-grained, comprehensive respiratory markers in real-time including inhale phase, exhale phase, breathing rate, breathing depth, and breathing symmetry to generate biofeedback on a companion mobile app for the user’s guided breathing exercises. Specifically, breathing monitoring is performed by sensing the head motion caused by respiratory effort using an accelerometer on the earbuds. The key novelty of BreatheBuddy lies in the fact that it is able to capture rich respiratory biomarkers in real-time using wireless commodity earbud. We show that our algorithms can track the inhalation phase with 90.91% F1-score and can estimate breathing rate with 95.05% accuracy. We implement our algorithms on-device and further show one breathing exercise user experience design that integrates respiratory biomarkers to generate biofeedback on a smartphone. We envision that our algorithms presented in this paper will instigate novel breathing exercise user experience design and engage more earbud users in breathing exercises to handle ever-increasing stress in their daily life.

In summary, the key contributions of the paper are the following:

(1) Demonstrates the feasibility of tracking on-going breathing phases in real-time using wireless commodity earbuds
(2) Presents algorithms that automatically generate a comprehensive set of respiratory markers including breathing phase, rate, symmetry, and depth for tracking real-time breathing exercises

(3) Shows an example user experience design by implementing our algorithms in resource-constraint earbuds and the companion smartphone

2 RELATED WORKS

This section describes the existing breathing detection algorithms and breathing biofeedback technologies utilizing smartphones and wearable devices. We also highlight the differences between our approach and the existing methods.

2.1 Breathing Rate and Breathing Phase Tracking

Breathing Parameter Estimation. Breathing is a complex physiological process of inhaling oxygen from the air and exhaling carbon dioxide from the body. Breathing rate is one of the critical vital signs and can be extracted from other physiological signals, airflow, and respiratory-related body movements [25]. A sensor or device next to the mouth or nose is a common approach to estimating airflow/breathing [4]. Breathing-related body movements can also be easily extracted by using inertial measurement units (IMU) attached to chest bands [36, 38], head-mounted devices [18], and wrist-worn devices [19, 50]. Several of those works either focus on user-initiated breathing measurement [18] or work in constrained environments such as sleep [29, 50]. Previous works also explored contact-less breathing measurement with a motion capture system [48], a smartphone microphone [39], and analysis of high-frequency wireless signals [37]. All previous approaches mentioned above show the feasibility of actively or passively detecting breathing using mobile sensors and focusing on breathing rate estimation. In this work, we focus on the passive breathing phase detection, breathing rate, depth, and symmetry estimation in real-time using low-power accelerometer embedded in commodity earbuds.

Biofeedback Driven Breathing Exercises A class of smartphone and smart-wearable based breath detection focused on controlled breathing to provide biofeedback for meditation [13, 16, 46]. Breeze [46] used smartphone acoustic data to detect breathing phases (inhale, exhale) to provide biofeedback in a particularly structured breathing exercise (4-second inhale, 2-second pause, 4-second inhale, 2-second pause with pursed-lip breathing). MindfulWatch [16] shows the feasibility of tracking breathing phases using smartwatch gyroscope data towards generating biofeedback for a particular body posture and wrist position. However, they kept the real-time biofeedback as a future work. Another group of researchers worked on a smart necklace [13] that detects breathing patterns related to different affective valance and arousal to communicate the emotion to the users and their loved ones through audio, haptic, and visual representation. Moreover, JustBreath [32], CalmCommute [6], AmbientBreath [23] presented breathing biofeedback through haptic feedback embedded in a driver's seat or through virtual reality display or audio-visual guide to reduce driver's stress. These are one way interaction where the breathing guidance are provided, but do not assess whether the users are actually following the guidance. On the other hand, we develop algorithms to track comprehensive breathing markers in real-time including breathing phase, rate, and symmetry using a commodity earbud and present a user experience where the user get the breathing biofeedback while following the visual breathing guidance. We hope that our work will inspire mobile interaction designers to design compelling and engaging breathing exercise user experiences utilizing the comprehensive set of breathing parameters passively tracked by the commodity earbuds.

2.1.1 *Earbud Sensing and Health Applications.*

Earbud Sensor Based Vital Sign Monitoring. Recent works show the potentials of earbud sensing to estimate breathing rate and heart rate [10, 22, 26, 33, 40, 43]. For example, inertial sensors such as accelerometer and gyroscope are used to estimate breathing rate in [33, 43]. On the other hand, in-ear audio captured by the earbud's microphone is used to estimate both breathing rate [10, 22, 40] and heart rate [10]. Several of them focus on a particular environment such as sleep [40] or work with heavy breathing after exercise when the breathing sound becomes more audible [22]. These works establish the feasibility of monitoring both breathing rate and heart rate using the same earbud device. In contrast, our approach presents algorithms to detect real-time, fine-grained breathing biomarkers including breathing rate, breathing phase, depth, and symmetry.

Earbud Sensor Based Interactive Applications. Recently, earbuds are getting more attention as popular human sensing and intervention form-factor for various interactive health applications [12, 30, 31, 40, 42, 55]. For example, CoughBuddy [30] uses both earbud's accelerometer and the microphone to accurately detect coughs from the earbud users. Ren et al., [40] utilizes earbuds microphone for fine-grained sleep monitoring. EarBuddy [55] and EarRumble [42] introduced novel face interaction through earbud sound detection to control the earbud's volume settings. A few other works [31] presented algorithms to estimate users' heart rate and smartly recommend music to calm down their heart rate. These applications show the growing interest in earbud sensing and intervention towards making the earbuds more useful for the users. In the current paper, we are presenting another novel approach to enable the earbuds in tracking user's real-time breathing for breathing exercises towards managing daily stress.

3 METHODS

Fig. 1. Earbuds sensing. It shows the positioning of the 3-axis accelerometer sensor in the earbud circuit board (left). It also shows the recommended orientation of the device in the ear (right).

3.1 Breathing Sensing Using Earbuds

Earbuds are becoming a popular form factor for multi-modal physiological sensing. Typically, earbuds have inertial sensors and microphones. Figure 1 highlights the inertial sensor inside our earbud's circuitry and the earbud's placement in the ear. It shows that usually X-axis of the accelerometer points parallel to the gravitational force. Therefore, when the user is sitting upright for breathing exercises, the breathing motion will be usually the up and down motion of the head. It will be captured by the X-axis of the accelerometer. This is expected because when we breathe in, the lung expands which moves the shoulder and the head upward. When we breathe out, the lung contracts which moves the shoulder and the head downward. This is the working principle behind our algorithm and the system design. Previous works [33, 43] show the feasibility

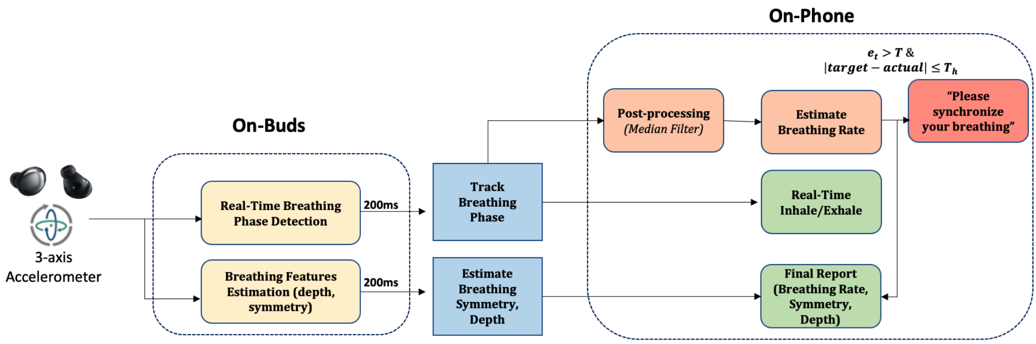

Fig. 2. System Overview. It shows how the real-time breath tracking works in earbuds and companion smartphone app to detect, analyze, and interact with the users.

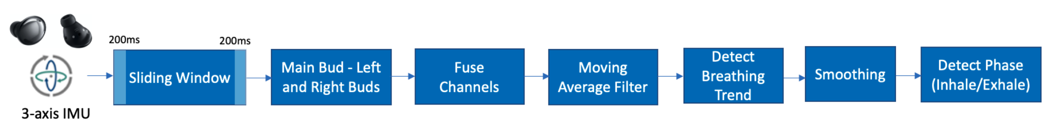

Fig. 3. Algorithm to track the real-time breathing phases running on the earbuds.

of estimating breathing rate minute by minute using earbuds. However, this work shows that earbuds can precisely track the on-going breathing phases (inhalation and exhalation) in real-time to generate a rich set of breathing parameters to design novel user experiences with biofeedback for breathing exercises.

3.2 System Design

Our earbud-based breathing exercise tracking system has two components - (1) algorithms running on the resource-constrained earbuds in real-time and (2) algorithms related to the biofeedback presentation and the user interactions running on the companion smartphone app. The system design overview is shown in Figure 2.

3.2.1 On-Bud Algorithms. On-bud algorithms have two branches. First, it tracks the real-time, ongoing breathing phases. Second, it estimates a comprehensive set of breathing biomarkers such as breathing rate, symmetry, and depth. This set of breathing parameters are sent to the companion phone app through a Bluetooth Low Energy (BLE) connection. These computations are running on the earbuds every 200 millisecond. Each sample is timestamped on the device so that they can be aligned in a time sequence on the phone.

Real-time On-going Phase Tracking. The details of the real-time on-going breathing phase tracking (inhalation or exhalation) are shown details in Figure 3. We use a sliding window with a step size of 200ms to determine the breathing phase every 200ms. Our algorithm first determines the main buds whether it is left or right and adapts the motion direction based on the main bud. It is because the left and right buds have orientation differences (Figure 1). We then fuse the 3-axis accelerometer signal. Based on the posture during the breathing exercise (sitting upright or lying on the back) either X-axis or the Y-axis carry the most information of the earbud motion related to breathing exercises. The Z-axis component of the breathing motion is negligible since it is perpendicular to the gravitation axis.

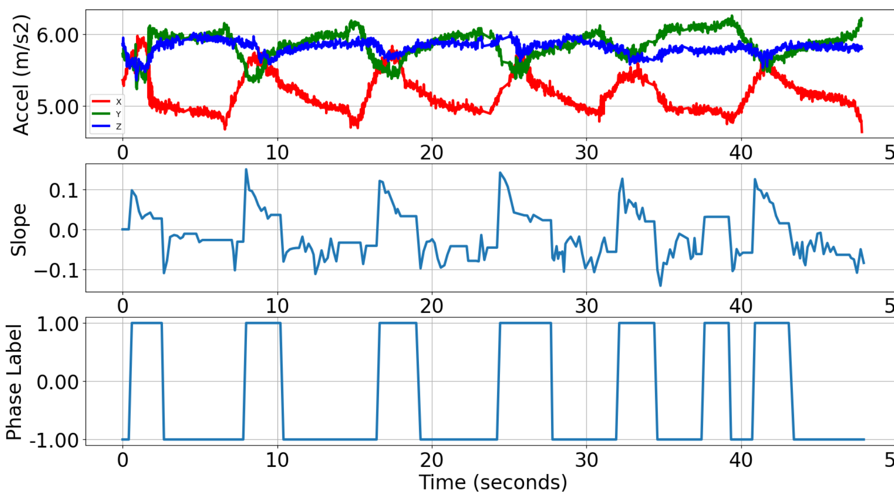

Fig. 4. Breathing phase tracking in real-time. (Top) Raw 3-axis accelerometer from 6 BPM exercise session. (Middle) Derivative of the breathing trend signal. Positive value indicates one phase and negative value indicates the opposite phase. (Bottom) Final phase detection output where +1 value indicates inhalation and -1 value indicates exhalation.

Figure 4 shows the raw accelerometer data during a breathing exercise session. This is from the left earbud while the subject is sitting upright. The subject was performing a symmetrical breathing exercise with 6 seconds inhale and 6 seconds exhale by sitting upright. In this posture, the X-axis clearly captures the breathing trend - upward trend during the inhalation phase and downward trend during the exhalation phase. We apply a rolling moving average filter with one-second window size to smooth out the raw signal and better extract the breathing trend. We then take the first difference of the signal to compute the breathing trend (second row of the Figure 4) using the following equation.

$$Slope_t = S_{t+1} - S_t \tag{1}$$

where $Slope_t$ is the derived slope at time t from the derived breathing waveform signal S.

Therefore, the upward trend signal generates the positive slopes (values above zero line) and the downward trend generates the negative slopes (values below the zero line). We then transform all the positive values as +1 and negative values as -1 using the following equation.

$$Phase_t = \begin{cases} -1, & \text{if } Slope_t < 0 \\ +1, & \text{if } Slope_t > 0 \end{cases} \tag{2}$$

where $Phase_t$ is the inferred phase label at time t based on the $Slope_t$ at time t.

However, due to occasional non-breathing head motion, false spikes are observed. Finally, we perform smoothing to reduce the false detection. We finally assign inhalation label to all +1 values and assign exhalation label to all -1 values. We then compare the predicted inhalation and exhalation phases timestamps with the annotated inhalation and exhalation timestamps to evaluate the algorithm performance.

Breathing Depth Estimation: Breathing depth is an important metric to distinguish a deep breathing from a shallow breathing. Usually, breathing depth is an indicator of how much air is inhaled into the lung while someone is breathing. Inhaled air is proportional to the lung expansion and lung expansion is proportional to the breathing head motion. Since our algorithm is designed to capture the breathing head motion, the amplitude of the extracted breathing signal becomes proportional to the actual breathing depth, especially during the breathing exercises.

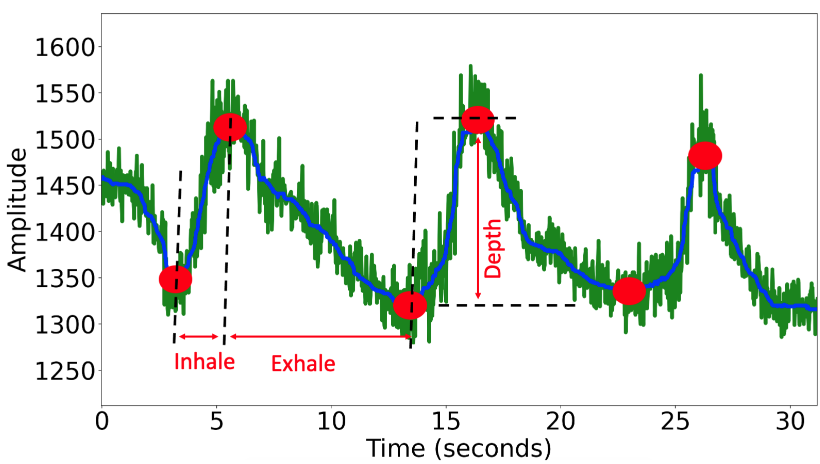

Fig. 5. Breathing Depth and Symmetry Computation. Depth is the amplitude difference between the peak and valley. Symmetry is the ratio between the inhalation and exhalation duration.

Our algorithm extracts the breathing waveform by removing the high-frequency components with a median filter (blue line in Figure 5). We then detect peaks and valleys by following the breathing peak detection algorithm described in rConverse [7]. Peaks and valleys are shown as the red dots in Figure 5. We then derive the breathing depth as the amplitude difference from the peak to the valley. To make the depth information more meaningful for the user, we normalize it with respect to maximum depth extracted from our dataset consisting of all the breathing exercises from 30 participants. Therefore, the depth information is presented as percentage (%). It indicates how deep is the breathing in the current exercise compared to other's maximum depth of breathing.

Breathing Symmetry Estimation: When we detect the peaks and valleys in the extracted breathing waveform shown in Figure 5, we compute inhalation duration as the temporal distance from the peak to the previous valley and exhalation duration as the temporal distance from the valley to the previous peak. We then take the ratio between the inhalation duration and the exhalation duration within the same breathing cycle as the breathing symmetry. Breathing symmetry is an important parameter to track since target breathing exercises can be symmetrical or asymmetrical. If the user is targeting asymmetrical breathing exercises (shorter inhale and longer exhale), the breathing symmetry will be lower than one. If the user is targeting symmetrical breathing exercises (inhale and exhale have the same duration), the breathing symmetry will be close to one. It can provide

(a) Users choose the target breathing exercise. **(b)** Feedback for on-going inhalation. **(c)** Feedback for on-going exhalation. **(d)** Feedback on breathing performance.

Fig. 6. User Facing App to Guide Users' Breathing Exercises.

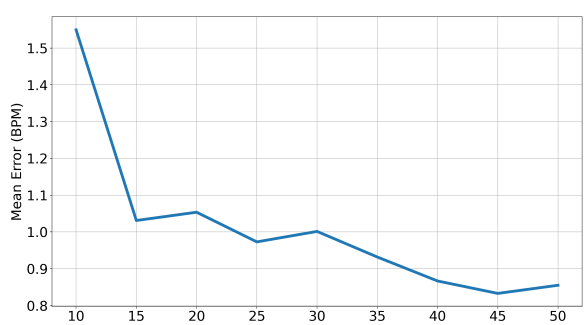

Fig. 7. Breathing Rate Estimation Error vs Analytical Window Size.

rich information about user's breathing pattern for better understanding of their performance compared to the target breathing pattern.

3.2.2 On-Phone Algorithms. On the other hand, on-phone algorithms process the timestamped breathing phase, depth, and symmetry data generated by the on-bud algorithms to produce users' breathing performance in real-time to understand whether they are following the target breathing pattern. The phone app has a visual guidance based on the target breathing exercise set at the beginning of the exercise session. The phone app generates the animation dynamically customized for both the inhalation and exhalation duration (Figure 6(a)) to guide the users to follow the target breathing pattern (phase duration, breathing rate, and symmetry). Our algorithms estimate the breathing rate to determine whether the actual breathing rate is deviated from the target rate. The phone generates feedback to help user synchronize their breathing as close as the target breathing. At the same time, the phone app shows the ongoing breathing phase (phase tracking information is received from the earbud) on the phone screen to highlight the differences between the expected and the actual phase. In the end, the app highlights their target breathing pattern and the actual breathing pattern (Figure 6(d)). Thus, the user can self-adjust their target breathing pattern for more effective breathing exercises.

Breathing Rate Estimation: When the phone receives the phase information for every 200ms window, it starts buffering the phase data with the timestamps while showing the phase detection output on the phone screen in real-time. As long as 10 seconds worth of data becomes available on the phone, a breathing rate estimation algorithm on the phone is triggered. Our algorithm first filters high-frequency components by using a median filter with a 3-second window. We then apply a modified zero-crossing algorithm to identify crossings only for the positive trend (Equation 1). We perform this since inhalation is a less noisy motion signal and accurately estimate breathing rate. We then convert the breathing rate (BR) from transforming the median time difference between two zero crossings into breaths per minute using the formula presented in [33].

$$BR = \frac{T}{T_w} \sum_{t=0}^{T_w} \mathbf{1}[S_t > 0 \land S_{t-1} < 0] \tag{3}$$

where $\mathbf{1}[.]$ is the identity function, S is the derived breathing waveform, T_w window size, and $T = 60s$ for breaths per minute.

To find the optimal window size for the breathing rate estimation, we compute the mean absolute error with respect to the window size. As shown in Figure 7, the error rate reduces (accuracy increases) with the increase of the window size. Our phone algorithm determines the actual breathing rate and computes the deviation of the actual breathing rate from the target breathing rate. If the actual breathing is too far from the target breathing (deviation > 2 BPM), then the phone app provides feedback as an alarm to the users that they need to synchronize their breathing with the target breathing rate. Therefore, the breathing rate estimation needs to be very accurate. The Figure 7 shows that our algorithm can start estimating a highly accurate breathing rate (error <1 BPM) from 25 seconds. Therefore, in the app, we show the breathing rate deviation alarm starting from 25 seconds to reduce the user burden with the false alarms.

Breathing Exercise Performance Assessment: Finally, we show the overall breathing performance at the end of the breathing session. It highlights the target breathing rate and actual breathing rate, target breathing symmetry and the actual breathing symmetry on the line charts. Breathing depth is shown in a pie-chart to show how closely they can achieve the target breathing depth (100%). It provides self-descriptive performance reports for the users to understand their performance and find the better breathing exercises based on their performance.

4 DATA COLLECTION STUDIES

We have collected data in two phases. We first conduct a study with 30 subjects to collect breathing exercise data using Samsung Galaxy earbuds for developing the algorithms. We then implement the algorithms on-bud and on-phone, design a breathing exercise biofeedback user experience, and then evaluate the system with the six independent quality assurance (QA) team members.

4.1 Data Collection for Algorithm Development

We have conducted the first study with 30 participants (15 male and 15 female, age range: 22-58 years) in a lab environment, where we asked our participants to wear the earbuds during the data collection. The participants are self-reported healthy adults and not having any cardio-respiratory issues which might interfere with breathing exercise activities. Seven of them did not have any experience in using the earbuds prior to the study. The remaining 23 participants were having experience in using the earbuds mainly for phone calls and listening to music. The participants were asked to perform breathing at their regular pace in a sitting position for 1.5 minutes. The duration of the subsequent breathing tasks is one minute each. The participants were performing slow-paced breathing with 5 breaths per minute (BPM) and 10 BPM. Previous studies [46] on developing guided breathing technologies inspired us to choose breathing duration and pace for each breathing session in our study. The participants were shown a visual animation to help follow the respective breathing pace. Our study was conducted during COVID-19; therefore, the participants wear a mask to ensure the safety of other participants and researchers. The study was approved by Institutional Review Board.

We develop an android app running on the Samsung Galaxy S9 smartphone to collect 6-axis inertial sensor data (3-axis accelerometer and 3-axis gyroscope) from Samsung Galaxy Buds Pro earbuds at 50Hz sampling frequency. We have also collected earbuds audio with a 16KHz sampling frequency. We use Zephyr Bioharness Biomodule3 as a reference device to derive the breathing waveform and reference breathing rate. The phone works as a data collection hub from the devices and sends data to the server via WiFi.

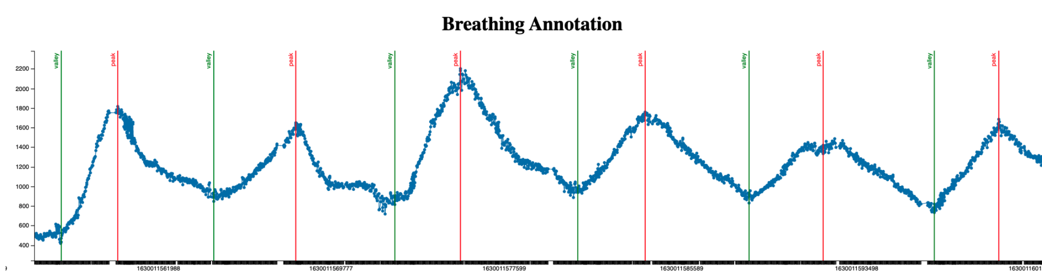

Fig. 8. Annotation Interface. Expert annotators annotated the peak and valley of the breathing signal to determine the inhalation phase (valley to peak) and the exhalation phase (peak to valley). X-axis indicates timestamps and y-axis represents the signal amplitude. Peaks positions are marked by red vertical line and the valley positions are marked by the green vertical lines.

The reference chest band device provides a detailed breathing waveform at 18Hz and the summary breathing rate at 1Hz frequency from a proprietary algorithm. Previous research [35] shows that summary breathing rate may not always be reliable as a reference breathing rate. Therefore, we develop a C3[11] based semi-automated, interactive interface to annotate each breathing cycle using the breathing waveform data with timestamps. We have implemented the state-of-the-art breathing cycle (peak and valley) detection algorithms on chest band breathing waveform presented in [7]. We load the breathing waveform data and the automatically detected breathing cycle peak-valley markers on the same interface (Figure 8). Researchers then visualize and adjust the automated detection if there is any missed peak/valley or false peak/valley. The reference breathing rate is calculated by taking the timestamp difference two consecutive peaks: $60/(ts_p^{i+1} - ts_p^i)$, where ts_p^i denotes the timestamp of i^{th} peak.

A subset of the dataset is annotated by more than one annotator to compute inter-rater agreement. If the signal quality is not good (e.g., too many peaks, missing most of the samples, breathing phases are not identifiable due to signal saturation [34]), the annotation for that particular data segment is skipped. Overall, the inter-rater agreement is computed using Cohen's Kappa ($\kappa = 0.83$).

4.2 User Experience App Design for the Breath Phase Tracking Algorithms

After the development of our algorithms offline using the data collected from the first study, we implement the algorithms on-bud and on-phone (Figure 6). We design a complete end-to-end breathing exercise user experience utilizing the estimated breathing biomarkers to generate the biofeedback in real-time. The app design is inspired by a popular guided breathing app that is reviewed by 1.3 million real users [17].

The app has the following four user interfaces (UI):

(1) *User Instruction UI* explains the device placement, positioning, posture, and how to breathe during the exercises. We have mentioned that the user expected to sit upright, wear the earbud in the ear as shown in Figure 1 (right), and follow the breathing guidance animation.
(2) *Phase Selection UI* allows the user to choose the duration of the inhalation and exhalation phases (e.g., 3 second inhale, 5 second exhale). Slow paced breathing exercises are more suitable for stress relaxation [17]. Hence, we allow the user to set the target phase duration from 3 seconds up to 10 seconds in the app which correspond to 3 breaths per minute to 10 breaths per minute (see Figure 6a).
(3) *Breathing Guidance with Biofeedback UI* shows intuitive inhale and exhale animation along with the algorithm detected breathing phases as the biofeedback (see Figure 6b, 6c). Growing flower indicates the users to inhale the air into the lung which resembles the lung and the chest expansion during the inhalation phase. Shrinking flower indicates the users to exhale the air outside the lung which resembles the chest contraction during the exhalation phase. This flower motion animates the breathing phases, the speed of the animation is correlated with the breathing rate, the size of the expansion indicates the breathing depth, and the relativity of the inhalation-exhalation speed indicates the breathing symmetry. Moreover, we also present the algorithm detected breathing phase label below the flower animation by highlighting it using a red underline.
(4) *Performance Feedback UI* shows the target breathing and actual breathing rate, depth, and symmetry at the end of the breathing exercise session (see Figure 6d). This can help realize their breathing exercise performance. If their breathing rate or depth is far from the target, they may set a different and more comfortable breathing exercise in their next exercise session. This can also help track the user breathing performance improvement over time for the repeated exercises.

4.3 Real-Time Evaluation with Quality Assurance (QA) Testers

We then give our end-to-end system to the quality assurance (QA) team to evaluate the user experience. Six participants between 25 to 40 years of age including five male and one female performed the evaluation test. Two of them did not have any prior experience using the breathing exercise applications. Each of them performed two different breathing exercise sessions including both symmetrical (5 seconds inhalation, 5 seconds exhalation) and asymmetrical (3 seconds inhalation, 5 seconds exhalations) breathing. Each breathing exercise session was one minute long. To evaluate the performance in more realistic scenario, the test participants were performing the breathing exercises independently at home, without wearing the face mask and the chest band.

The participants have recorded the phone screen video and our phone app has recorded the raw sensor data, and the derived data while the participants performed the on-device tests in real-time. Two researchers have independently coded the videos to match the actual breathing phases with the detected breathing phases and estimated approximate latency. We then take the average latency between the two raters.

Breathing Frequency	Breathing Phase	Accuracy
5 BPM	6 sec inhale, 6 sec exhale	84.4%
10 BPM	3 sec inhale, 3 sec exhale	86.94%

Table 1. Breathing phase tracking accuracy (offline)

5 RESULTS

5.1 Phase Tracking Evaluation

We have applied our algorithm to the collected breathing exercise datasets from the 30 subjects to compute the performance. For the performance metric, we report the accuracy which is computed from the confusion metric based on each 200ms output and the annotated phases.

We observe that our algorithm can detect the phase with 84.4% accuracy on the 5BPM dataset and 86.94% accuracy on the 10 BPM dataset (Table 1). We observe slightly better accuracy on the 10 BPM because this exercise was relatively easier to perform for all the participants and this is closer to their baseline breathing rate. This is a good accuracy compared to the state-of-the-art breathing phase-detection model Breeze [46] which is detecting breathing phases using acoustic data to provide breathing biofeedback during meditations. Our phase detection accuracy is 9-11% higher just using a low-power accelerometer. However, since this evaluation is on the offline dataset, it is not possible to estimate the latency between the on-bud phase detection and on-phone feedback visualization.

Hence, we have also evaluated the real-time breathing phase detection with latency performance on six independent QA testers. The results are shown in Table 2. Overall, average phase detection accuracy is 92.40%, average precision is 87.25%, average recall is 94.96%, and F1-score is 90.91% with average latency is 1 second. The latency is explained by the fact that we have a 200ms window size, 100ms on-bud processing time, 500ms average delay for the rolling average window, communication overhead, and the operating system's context switching time. We observe higher test accuracy is because the test users were using the end-to-end app with the algorithm detected biofeedback. We hypothesize that the user experience design and the real-time breathing biofeedback generated by our algorithms might have helped improve the breathing performance compared to the first study. However, it warrants a future study to test our hypothesis.

Among all the test subjects, P4 was having relatively lower accuracy because the target breathing rate was difficult for this subject to perform. It is perhaps due to lower lung capacity. It warrants future studies to establish the relationship between lung function with the optimal breathing exercise pattern.

5.2 Breathing Rate Evaluation

The breathing rate estimated for tracking breathing exercises needs to be very accurate since it provides an alarm to the users on their breathing performance. Lower accuracy will cause a lot of false alarms and may eventually lead to lower engagement.

We have evaluated the breathing rate algorithm in the context of both symmetrical and asymmetrical breathing to understand whether there are any performance differences between them. We consider mean absolute error (MAE) and mean absolute percentage error (MAPE) as the metrics for the breathing rate performance evaluation. We observe that the performance is similar in both symmetrical and asymmetrical breathing. The overall breathing rate estimation error (MAE) is only 0.31 BPM with the average percentage error (MAPE) is only 4.95%. It shows the generalizability of our algorithm across both symmetrical and asymmetrical breathing.

Subject	Accuracy	Precision	Recall	F1-score	Latency
P1	97.71%	94.44%	98.07%	96.23%	0.5
P2	89.75%	84.78%	91.76%	88.13%	1
P3	94.38%	90.24%	97.36%	93.67%	0.5
P4	84.36%	79.24%	88.42%	83.58%	1.5
P5	92.91%	82.69%	97.72%	89.58%	1
P6	95.31%	92.13%	96.47%	94.25%	2
Average	92.40%	87.25%	94.96%	90.91%	1

Table 2. Real-time on-going breathing phase tracking accuracy, precision, recall, F1-score and average latency in seconds (on-device).

Breathing Type	MAE	MAPE
Symmetrical Breathing	0.35 ± 0.37 BPM	6.05 ± 7.85%
Asymmetrical Breathing	0.29 ± 0.39 BPM	4.35 ± 3.97%
Symmetrical + Asymmetrical	0.31 ± 0.39 BPM	4.95 ± 6.86%

Table 3. Breathing rate estimation accuracy. MAE = Mean Absolute Error. MAPE = Mean Absolute Percentage Error.

Subject	Breathing Depth	Breathing Symmetry
P1	87.63%	1.05
P2	89.21%	1.07
P3	85.02%	0.89
P4	85.20%	1.27
P5	88.07%	1.57
P6	92.98%	1.45
Overall	88.01%	1.21

Table 4. Breathing depth and symmetry from the on-device evaluation tests.

5.3 Breathing Depth and Symmetry Evaluation

We have also computed the breathing depth and the symmetry from the on-device tests in Table 4. The reference chest-band device Zephyr Bioharness Biomodule3 does not readily provide the breathing depth and symmetry data. Therefore, we qualitatively evaluate the breathing depth and breathing symmetry. We observe from the test performance that the average breathing depth was 88.01%. It indicates that the test subjects were performing the deep breathing exercises as the shallow breathing usually would fall around 50% boundary. The average symmetry is estimated as 1.21.

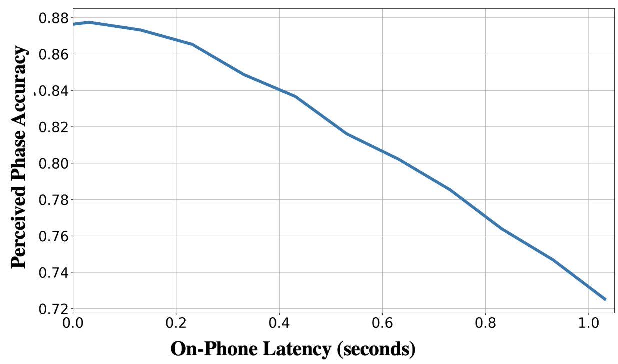

Fig. 9. The accuracy drop due to latency in phase detection on-bud and visualization on-phone.

6 DISCUSSION

6.1 Real-time Phase Detection Accuracy, Latency, and Perceived Accuracy on the User Interface

We mentioned in the result section that there is an average one second delay from the on-bud phase detection to on-phone phase visualization. One adverse effect of this latency is that the user may perceive higher inaccuracy while looking at the visual presentation of the phase information on the phone screen. It is because they know the actual start of their breathing phases (onset of their inhalation or exhalation), but the phone may receive phase information after one second of the actual onset. We quantify the effect of the delay on the accuracy which is termed as the user perceived accuracy on the smartphone interface. Perceived accuracy is calculated by shifting the window based on the delay. Figure 9 shows that accuracy decreases with the increase in latency. Actual phase detection accuracy of 88% on-buds can be perceived as 72% when the delay between the on-bud detection and on-phone visualization is around one second.

The latency is explained by window size, processing time, signal filtering techniques, communication overhead, and the operating system's context switches. User experience design should keep this in mind that there will always have some inaccuracy in the detection algorithm and there will be a delay between the phone and bud. In our design, we visualize the inhale and exhale phase transition information as a fading colors to handle the delay and the inaccuracies so that the user perception of the inaccuracies can be more fuzzy. This can be an interesting problem for the user interaction designers to design the real-time user experience on top of noisy inferences.

6.2 Earbuds for Stress Detection and Relaxation

Recent studies show that the earbud can be a suitable form factor for sensing multi-modal physiological data including heart rate [10] and breathing rate [35, 43] which can be used to reliably assess psycho-social stress [21]. Breathing exercise meditations are known to be effective stress relaxation methods for decades [28]. However, performing the breathing exercises in wrong ways can increase the stress further [3]. Our algorithms presented in this paper may enable the earbuds to become the 'one-stop' service device for both stress detection and stress relaxation.

6.3 Limitations and Future Works

There are several limitations of our work. First, it may be susceptible to non-breathing head motion. The accuracy may become lower if someone moves the head out of sync from the breathing motion, although this is not expected in breathing exercises. Future works can integrate audio fusion algorithms to handle the non-breathing head motion assuming that breathing sounds will be audible despite out-of-sync head motion. Second, we have experimented the breathing exercises without any breath-holding between the inhalation and exhalation phases. There are popular breathing exercises which include breath-holding. It needs future studies to evaluate how the breath-hold may affect our algorithm performance and the user experience design. Third, each of our experimental sessions lasted one minute. We hypothesize that our algorithm will perform with similar accuracy for longer duration since it detects breathing head motion. However, breathing depth may reduce over time for longer breathing exercises and the algorithm may become less accurate. To know the duration limit of our algorithms, we need to design future studies with longer duration exercises. Fourth, we did not perform the quantitative evaluation of the depth and symmetry estimation. Future studies with devices measuring tidal volume, minute ventilation, or more comprehensive thoraco-abdominal pressure during breathing can quantify these markers. The reference device can include smart-clothing such as HexoSkin[20] or the GoSpiro in relaxed mode [27].

7 CONCLUSION

In this paper, we present BreatheBuddy, a passive respiratory sensing system that monitors comprehensive breathing biomarkers including breathing phase, rate, depth, and symmetry during breathing exercises in real-time using earbud's accelerometer. Our algorithms work for both symmetrical and asymmetrical breathing exercises. Our evaluation with independent test users shows that it can detect breathing phases with 90.91% F1-score and estimate breathing rate with 0.31 BPM mean absolute error (95.05% accuracy). Our algorithms are tuned towards designing seamless breathing exercise user experiences given the inaccuracies in the detection algorithms. It facilitates real-time breathing biofeedback to potentially make earbud as an effective tool for breathing exercises towards stress relaxation.

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